

Module 05 Lab: Data Preparation

Course: ITAI 1371

Date: September 19, 2026

Min:

This lab divide the process of data preparation into several logical stages like data cleaning, feature transformation and running the lab was similar to the approach used in a real data science project since having many minds working together in understanding the details of the task enabled us to discover issues like the fact that Pandas outputs True or False instead of 1 or 0 during One-Hot Encoding. The process has not only taught us how to write code but also clarified for us the basic idea of mathematical reasoning behind it, because we have consciously used median imputation in order to avoid distortion of the Age column by very extreme outliers, applied `drop_first=True` so that multicollinearity and dummy trap are avoided without a loss of baseline information and implemented `StandardScaler` to prevent distance-based algorithms from wrongly interpreting very high Fare numbers. Overall, the lab illustrates the Garbage In, Garbage Out concept, because training of models is considered to be glamorous time-consuming creative task; however, it is not widely known that without lengthy preparation of data, algorithms would not be able to identify useful trends in raw data information like the Titanic dataset.

Camille: This assignment gave me a better understanding of how important data preparation is in machine learning. I learned that the dataset contained 177 missing Age values that needed to be replaced using the median age. It showed me how missing information can affect the quality and reliability of a dataset. I also learned why choosing the appropriate method for handling missing data is important. The lab made the overall process feel much less intimidating and easier to understand. I gained a better appreciation for the preparation required before machine learning can begin. Overall, this assignment strengthened my understanding of data cleaning and its importance.

Chris:

Working with the Titanic data set helped me to understand the importance of the data preparation in the machine learning process. The data set provided is not ready to use in the majority of the cases due to missing values, categorical data and different scales of the numerical data. The main part that I did for the assigned task of Task 3 was the use of the `'StandardScaler'` to standardize the values in the `'Age'` and `'Fare'` columns. To test the part that I worked on I created a copy of

the DataFrame where I temporarily completed the one-hot encoding for the categorical columns in the data set. From the original values in the `Age` and `Fare` columns to the standardized values in the same columns I learned what feature scaling does. I learned that the median can be a good option for replacing missing values in the `Age` column because the median is less affected by outliers than the mean. I also learned that the one-hot encoding of the categorical features in the data set converts the categorical values into numerical values that can be used by machine learning algorithms. Finally, I learned that feature scaling is not always necessary. For example, Decision Trees split the features based on the feature values themselves and not based on the distances between the feature values. This lab gave me a lot of practical experience with the data preparation and how the missing-value handling, the encoding and the scaling can be used to prepare the data for the machine learning.

Samantha:

For this lab, my main contribution to the group was completing Question 1 of the Knowledge Check and helping with the final preparation and submission of our work. My question focused on understanding why the median can sometimes be a better choice than the mean when filling in missing values in a dataset. I learned that the median is less affected by extremely high or low values, also known as outliers. Because of this, using the median can give a better representation of the typical values in the data and prevent unusual values from affecting the results too much. I also helped with the final preparation of the assignment by reviewing the notebook and making sure that each group member's section was completed. This included checking that the required tasks and questions were answered and making sure the notebook was organized and ready to be submitted. Overall, this lab helped me understand that data preparation is an important part of machine learning because the quality and organization of the data can affect how well a model works. I also learned more about working as a team and how each person's contribution helps complete the overall project.

Rigoberto:

Through this assignment, I learned more about how categorical data needs to be prepared before it can be used in a machine learning model. I learned how one-hot encoding changes categories such as Sex and Embarked into numerical columns that a model can understand. I also learned why using `drop_first=True` helps avoid redundant information. At first, I had a `pd NameError` because the setup code had not been run, but I learned that the required libraries and setup need to be executed before running the encoding code. Overall, this task helped me better understand data preprocessing and how it connects to the later machine learning steps.